Paper Review | arXiv 2611.04527 | 2026-11-04

Future Representation Distillation for Imagination-free World Action Models

DreamFree: distilling VDM "foresight" into a single-frame encoder: better robot actions, no future-video generation

Jaehyeong Jo*, Sangwon Jang*, Sihyun Yu, Seonghyeon Ye, Yekyung Nah, Hojung Jung, Kimin Lee † , Sung Ju Hwang †
KAIST | Code: github.com/harryjo97/dreamfree

TL;DR

Gist: "imagine the future" via training-time features, not test-time pixels

A frozen video diffusion model (seeing the full future) as teacher → distilled into a video DiT encoder that sees only the current frame → **a single forward pass** yields future-aware features → an action DiT directly outputs action chunks.

64%
Real-world DROID avg. success
DreamZero 56% • π0.5 16%

11 Hz
With a 14B video backbone: real-time closed loop (91 ms/chunk)

17.8×
Inference speedup over video-generating WAM DreamZero

74.4%
RoboCasa-Kitchen, 24 tasks
14B + extra unlabeled videos

Bonus: Stage 1 needs no action labels, so new tasks can be learned from unlabeled videos (even from other robot embodiments).

Issue: video-gen WAMs are slow and easily misled by wrong futures

VLA (e.g., π0.5)	Video-generating WAM (e.g., DreamZero)	DreamFree (ours)
■ Image-text pretraining, strong semantic priors	■ Inherits predictive knowledge of video models	■ Predictive knowledge distilled into current-frame features
■ Single forward pass, fast	■ Iterative video denoising: ~1.6 s / chunk	■ No video generation at inference; single forward pass
■ Lacks spatiotemporal dynamics / physical evolution knowledge	■ Wrong future video prediction → wrong actions	■ Goal: VLA speed with WAM priors

Key question: Does action prediction really require pixel-level "imagination" of the future, as in video generation? Can predictive information appear directly in current-frame visual features?

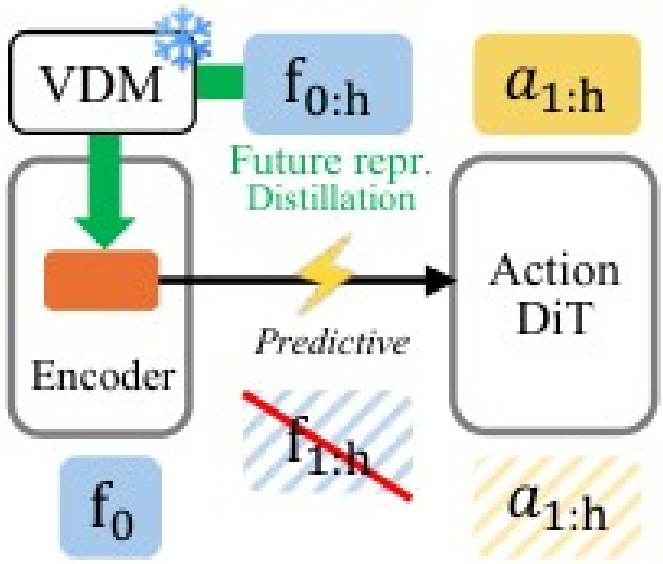
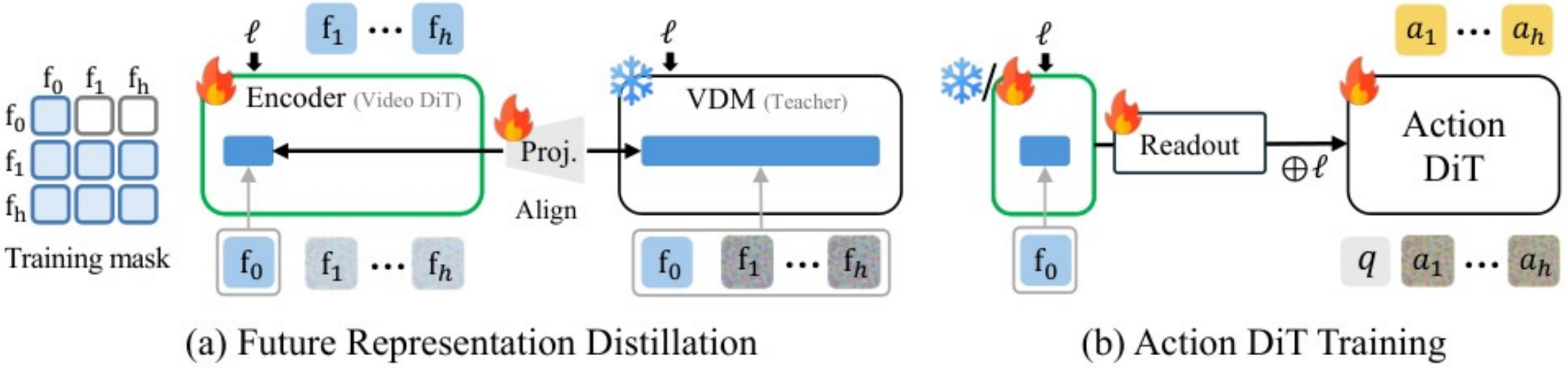


Figure 1 (left): VDM teacher distills into the encoder; future frames f1:h are no longer generated

Method: two-stage training: first a "future-aware" visual encoder, then actions



Stage 1 • Predictive Visual Encoder

Initialized from Wan2.2 video DiT; a structured mask lets the current frame attend only to itself; its hidden states are projected to align with the frozen VDM's future representations. No action labels.

Stage 2 • Action Module

Encoder frozen; a 64-query Perceiver readout extracts action tokens; a 1B action DiT, conditioned on instruction and proprioception, outputs 32-step action chunks via flow matching.

Inference • No Video Denoising

Encoder runs once at t=0, without CFG; actions use 10–16 Euler steps; the teacher VDM and projector are discarded at deployment.

Key: cosine-align a current-frame-only student to a future-seeing teacher

Training Objective

$$L_{distill} = -\mathrm{sim}(\, p_\psi(h_i),\ g_j(x_{td}, t_d, \ell)\,)$$

Projected student current-frame features ↔ frozen teacher's layer-j features of the noised full video

$$L_{gen} = \| \, u_\theta(x_t, t, \ell) - (\varepsilon - x^*) \, \|^2$$

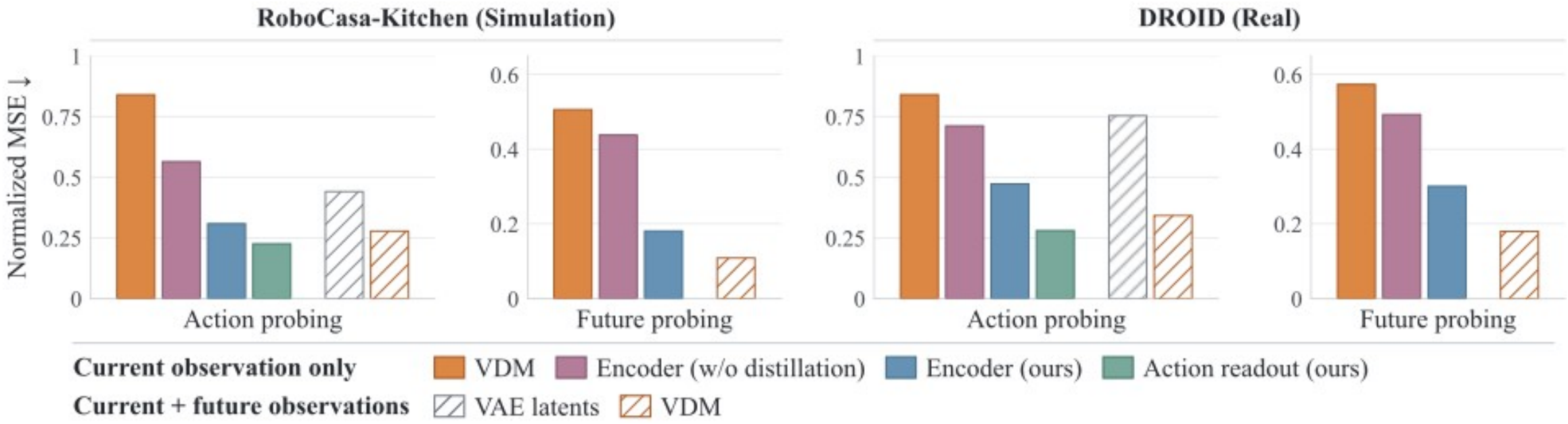
Native flow matching: clean current frame + noised future frames, preserving predictive ability

$$L = L_{gen} + L_{distill}$$

Both terms summed with equal weight

- **Structured attention:** current-frame tokens attend only to each other; each future frame sees only the current frame and itself, with no text, forcing planning information into the current-frame representation.
- **Projector:** the N current tokens are replicated into F×N, plus learnable frame embeddings, then a 2-layer decoder aligns them with the teacher.
- **Supervise future positions only:** current-frame tokens are excluded from the distillation loss; cosine distance after non-affine LayerNorm.
- **Hyperparameters:** teacher noise t_d = 0.7 (sim) / 0.8 (DROID); 14B backbone i = j = k = 31.
- **Training cost:** both DROID stages take ~1 week on 16 H200s.

Distilled current-frame features: more action/future-aware, near seeing it



Action Decodability

Encoder error < no-distill variant < original VDM; distillation gives the largest drop

Current Frame Only

Action error below VAE latents of "current + true future", near the future-seeing VDM

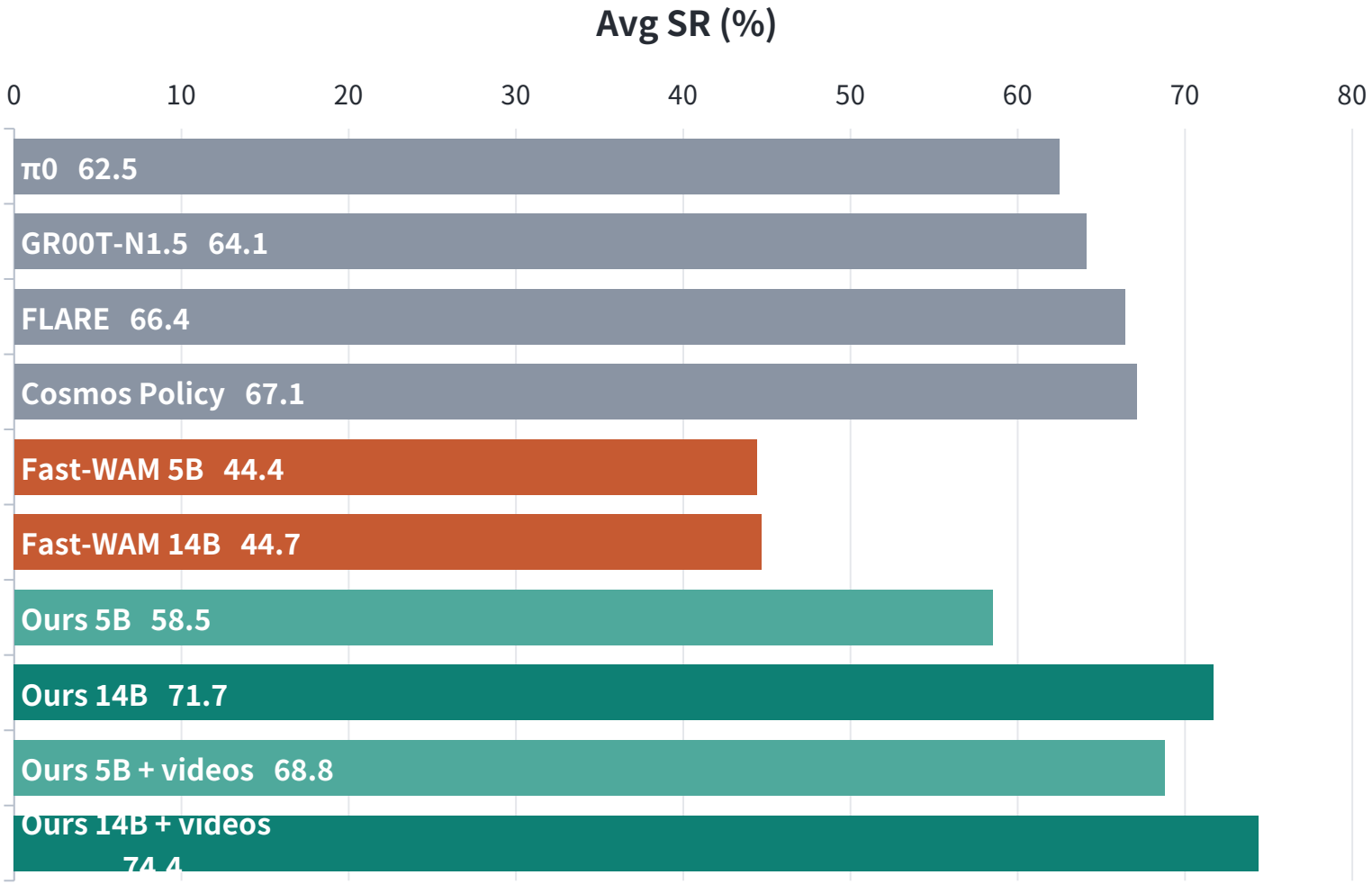
Readout Features

Error drops further; action-relevant info is effectively extracted

Limitation

On DROID, a clear gap remains between future probes and the future-seeing VDM

RoboCasa: 71.7% with 14B, 74.4% with unlabeled videos, far above Fast-WAM



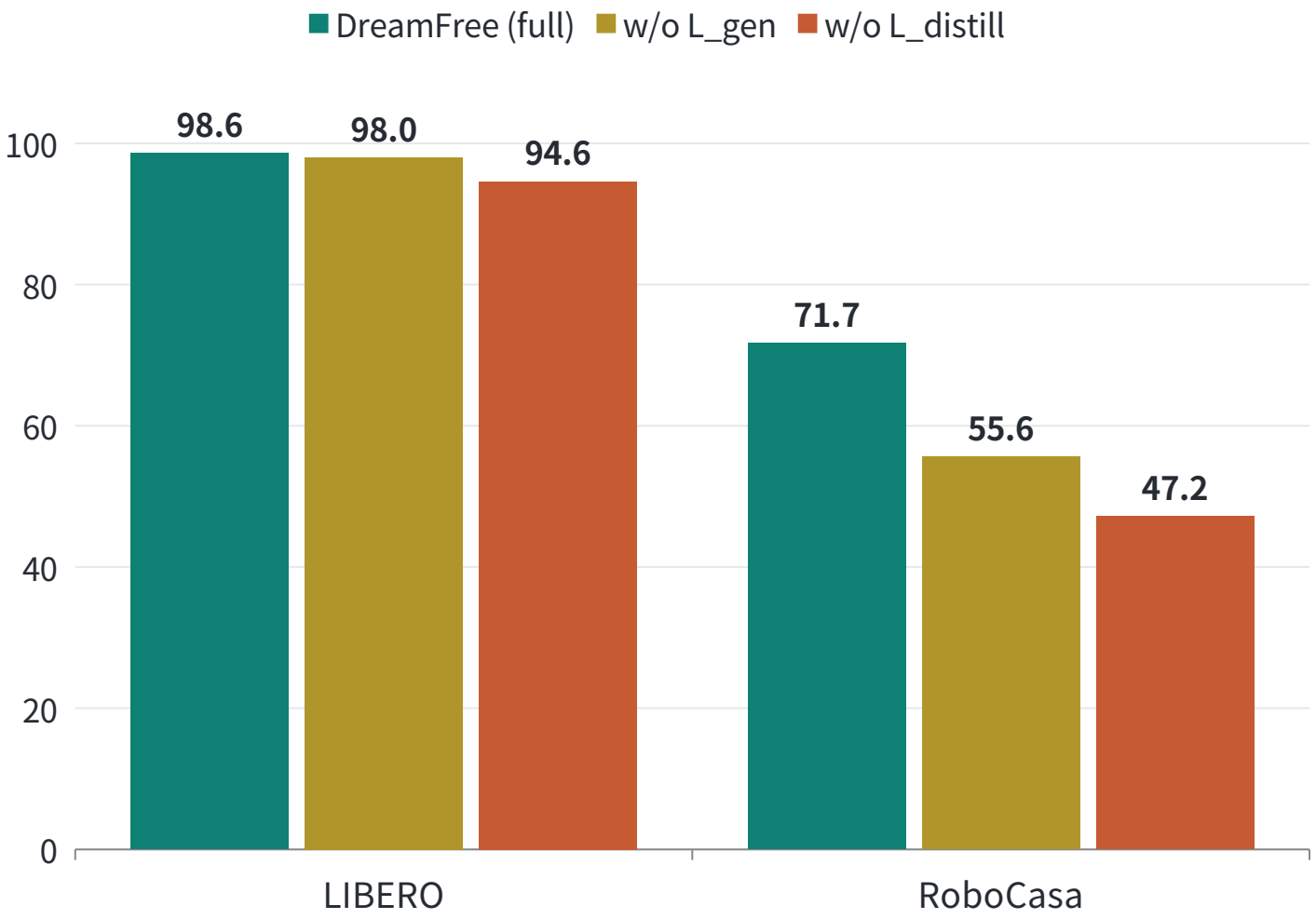
+13.2 pp
Gain from 5B → 14B backbone
(Fast-WAM only +0.3 pp)

+2.7 / +10.3
Gain from adding unlabeled videos
(14B / 5B, in pp)

98.6%
LIBERO avg. over four suites
(LingBot-VA 98.5, near saturation)

arXiv 2611.04527 | Source: Table 2 (avg. success % on 24 RoboCasa-Kitchen tasks), Table 1, Table 7

Ablation: no distill loss hurts most; RoboCasa 71.7% → 47.2%

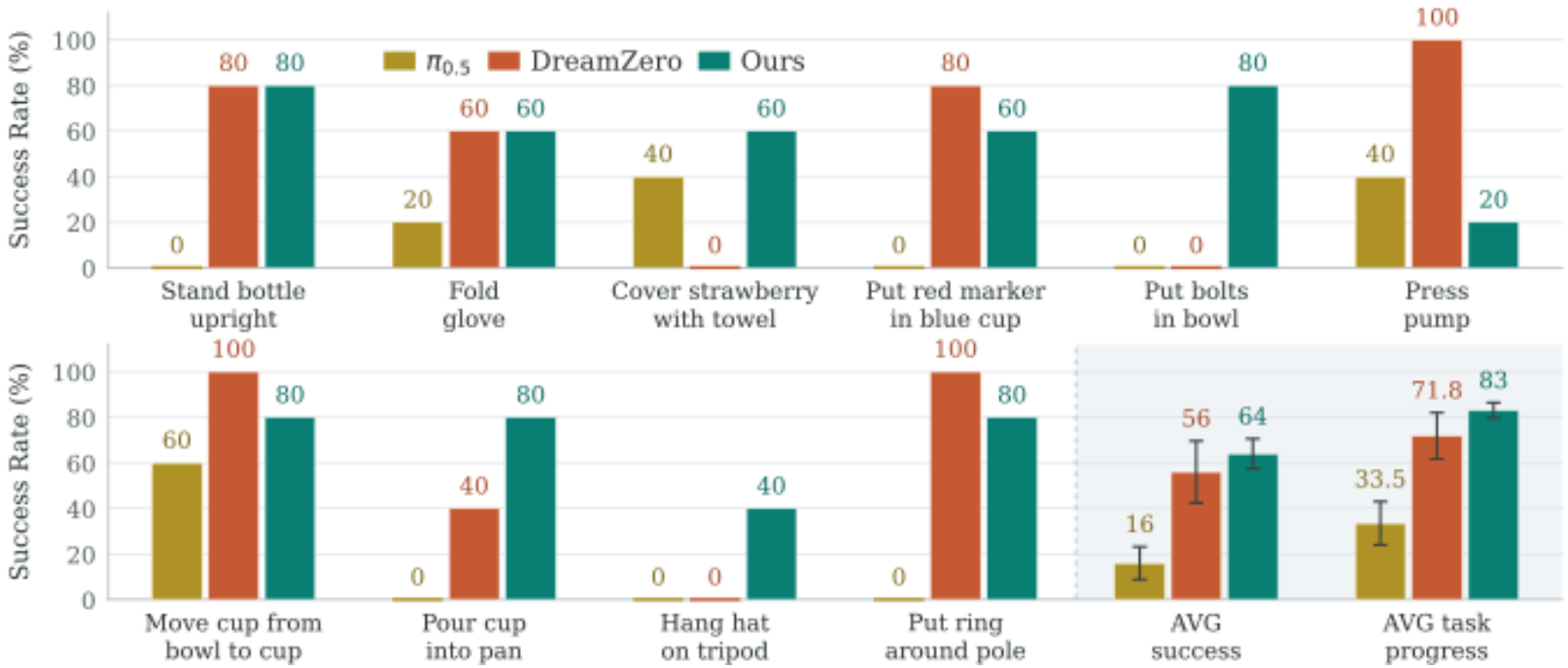


−24.5 pp
RoboCasa w/o L_distill

−16.1 pp
RoboCasa w/o L_gen

- Both objectives help; distillation dominates
- Consistent with probes: without distillation, both action and future decodability drop

Real DROID: 64% avg. success; best on deformable/multi-step rearranging



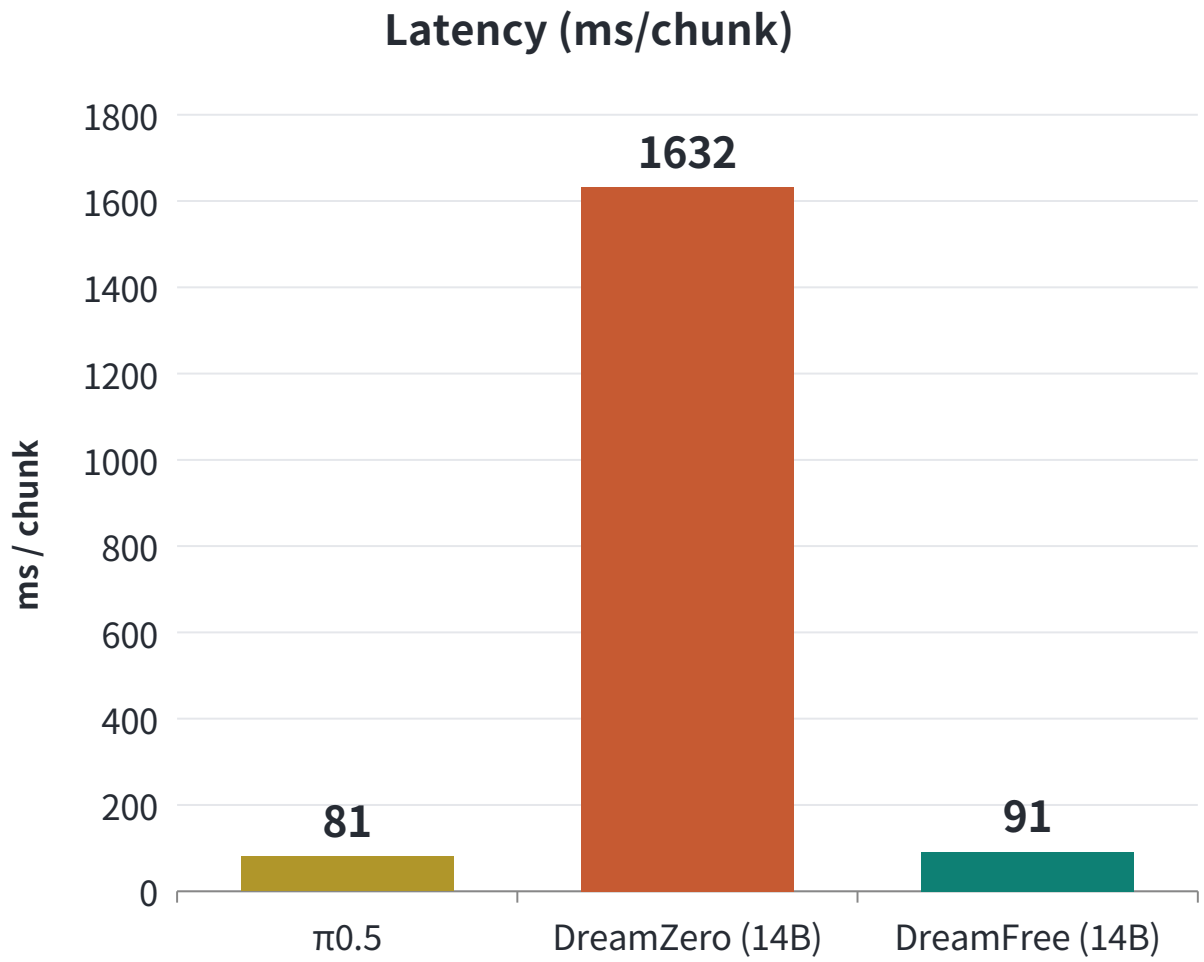
64%
Avg. success rate
DreamZero 56% • π0.5 16%

83%
Avg. task progress
DreamZero 71.8% • π0.5 33.5%

DreamZero 0% → ours:
Cover strawberry 60% • Place bolt 80% • Hang hat 40%

Still behind DreamZero:
Press pump 20% vs 100%; place red pen, cup transfer, ring placing

Efficiency: 91 ms / chunk, nearly as fast as VLAs, 17.8× faster than DreamZero



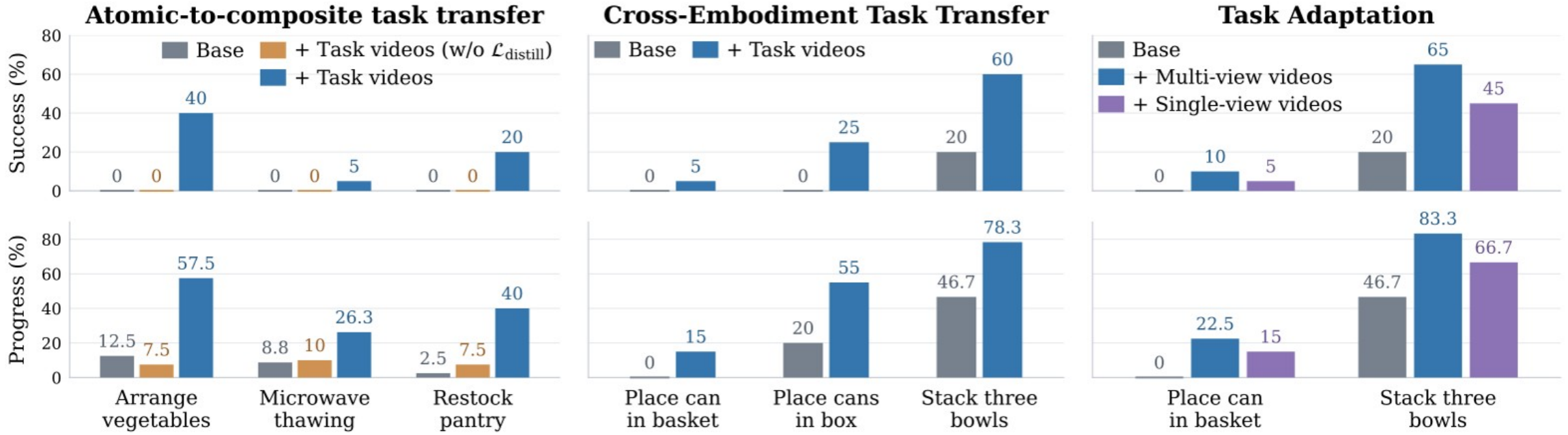
17.8×
faster than DreamZero

1.1×
of π0.5 latency

Why is it fast?

- **Algorithm:** single encoder pass (t=0); no video denoising, no CFG
- **Engineering:** VAE-encode current frame only, fused attention & QKV, torch.compile, CUDA graphs
- **Task time:** placing 6 blocks takes clearly less total time than π0.5 and DreamZero

Unlabeled video: compos. tasks 0→21.7%, cross-emb. 6.7→30%



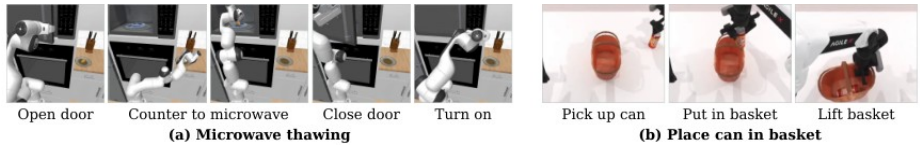
0 → 21.7%
Atomic → compositional (same embodiment)
still 0% without distillation

6.7 → 30%
Cross-embodiment: Franka bimanual videos
→ executed on ALOHA-AgileX

20 → 65%
Post-training: encoder tuned on 10 videos
stack three bowls (multi-view), ~10 min

Limitation
If the base policy cannot open cabinet doors, videos alone cannot teach "put into cabinet": videos add task knowledge, not new motor skills.

Qualitative: bimanual YAM; compositional tasks learned from video only



Compositional tasks learned only from videos: microwave thawing (4 subgoals) and basket lifting

- **Bimanual YAM:** ~100-hour subset of ABC-130k, no cross-embodiment pretraining, tested out of the box in new scenes
- **Note:** bimanual results are qualitative only, with no quantitative success rates

Discussion

Verdict: clear idea, solid evidence; limited real-world trials & transfer

Strengths

- Replaces inference-time pixel generation of the future with training-time feature supervision
- Probes + ablations form a full evidence chain: distillation → future info → action decodability
- Vs. Fast-WAM: with only the generative loss, a larger backbone barely helps; with distillation, both backbone scale and video data pay off
- Stage 1 needs no action labels, enabling new tasks from cross-embodiment videos

Caveats

- Only 5 real-world trials per task (50 total), so the 64% vs. 56% gap is not statistically robust; still behind DreamZero on pressing/insertion tasks
- Low absolute video-transfer success (21.7%, 30%); closer to proof of concept
- Three-stage training (backbone adaptation / distillation / action), ~1 week on 16×H200; layer indices chosen manually
- Part of the 17.8× speedup comes from engineering; extra videos and cross-embodiment pretraining untested on DROID

Take-away

The value of video models for robot control lies in their learned "predictive representations", not necessarily in actually generating future videos.

Method	Frozen VDM teacher → future representation distillation into a current-frame-only student encoder; single-pass inference
Results	DROID 64% success, 11 Hz; RoboCasa 74.4%; LIBERO 98.6%
Scaling	Larger backbones, more unlabeled videos → better control; new tasks from cross-embodiment videos

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